

Alberta Health SFTP External User Guide for GoAnywhere

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Alberta Health

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Document Revision History

Version	Version Date	Summary of Changes	Author
1.0		Initial Release	IBM
1.1	September 25, 2015	Updated to reflect changes as a result of the SFTP server migration.	CGI
1.2	June 28, 2016	Updated to include method to reset password	CGI
2.0	July 23, 2018	Overhaul to address support burden of Alberta Health business owners for external SFTP users.	CGI

Version	Version Date	Summary of Changes	Author
2.1	August 15, 2018	Made changes to 2.0 version requested by Alberta Health.	CGI
2.2	March 16, 2021	Minor updates and formatting revisions.	CGI
2.3	Sept 13, 2022	Update SSH key pair generation	IBM
2.4	Sept 26, 2022	Update info from CGI to IBM handling	IBM
2.5	Nov 7, 2022	Update business owner and service desk contact info	IBM
2.6	Nov 16, 2023	Update passphrase is required	IBM
2.7	Feb 16, 2024	Update stale file deletion period	IBM

Deliverable Approvers and Reviewers

Use the following table to name deliverable approvers and reviewers. Deliverable approvers and reviewers should be the same as those already stated in the Project Management Plan for this deliverable.

Title and Name	Type	Purpose
AH Production Management Dean Mayo, Dale Loroff	A	Confirms proper documentation is in place to enable ongoing support of the GA System.
AMS Application Team Karen Mah, Jim Borynec	R	Confirms proper documentation is in place to enable ongoing support of the GA System.
AMS DevOps Eva Li	P	Prepares work product

Type: **A** – Approves (implies review as well); **P** – Prepares; **R** – content review and input; **I** – Information Only

1 Document Purpose

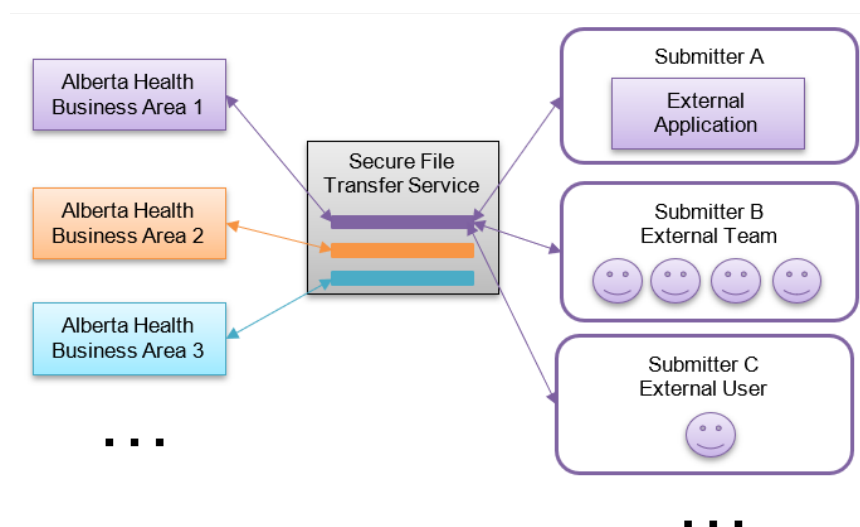
This document is a guide for users of Alberta Health's secure file transfer service that are external to Alberta Health. It is intended to be provided to an external user by an Alberta Health business owner as a guide to setting up and maintaining information exchange.

2 File exchange with Alberta Health

Alberta Health has established a secure file transfer service to enable the secure transfer of sensitive information between Alberta Health and other parties.

2.1 Relationships

The service manages all the relationships between particular business areas at Alberta Health, and external users or organizations, called “submitters”. (The common case is for Alberta Health to receive data from external parties, but submitters can also be consumers of data.) A submitter can be an application, a team, or an individual. In the case of the team, multiple user accounts will exist, and will all see the same files.



2.2 Exchange method and workflow

To implement the secure file transfer service, Alberta Health operates servers running a system called “GoAnywhere”. This name appears in emails sent by the system, and is used in some processes and documentation. The GoAnywhere system can be accessed with different technologies: website, SFTP(Secure File Transfer Protocol), etc.; however, for security reasons, only SFTP is available to external users.

A user using SFTP to access the Alberta Health Secure File Transfer Service will see a submitter area with the following folder structure:

- **UPLOAD**
 - This folder is where files going to Alberta Health must be put. After a short while, the system will remove the file from this folder and place it in another folder (not visible to you) where Alberta Health will be able to see it.
- **DOWNLOAD**
 - This folder is where files coming from Alberta Health will appear.

- **Important:** Files are automatically deleted from the DOWNLOAD folder after 30 calendar days. The secure file transfer service is not an archival service, and has limited storage capacity.
- Files can also be deleted manually – we recommend deleting the files after you have downloaded them and verified their contents.
- **HELP**
 - This folder contains a read-only text file with information on terms of use for the system. It is not used in day-to-day information exchange activity.



readme.txt

Each submitter has its own folder structure, and a corresponding folder structure exists for the Alberta Health side as well. As a user, you will only see your submitter area and not those of other submitters or of Alberta Health.

3 Responsibilities

3.1 Security and Privacy

The Alberta Health secure file transfer service enables security and privacy, but only in partnership with users and organizations. In accordance with the Health Information Act, authentication details such as passwords and private keys must be kept secret, or they can be used to impersonate a user. Computers must be kept secure and up-to-date, to protect authentication details and private information downloaded from the service. Compliance to all applicable laws and regulations is required.

3.2 Management of user accounts

3.2.1 Role of the Alberta Health business owner

User accounts are set up to access submitter areas in the secure file transfer service. The submitter area is associated with a corresponding Alberta Health business area. The Alberta Health business owner is your partner in user account management, as they must approve changes to user accounts that access information from their business area. Password resets are an exception to this general rule and do not need to involve the Alberta Health business area.

3.2.2 When to make user account changes

New user accounts can be created as needed as people join your team. The process for creating a new account is provided in section 4.3 Creating an SFTP account.

A user account should be terminated if the account is no longer required. This could be due to an employee leaving the organization (examples: transferred, resigned, terminated, retired, etc.). Also, this could occur if your organization no longer exchanges information with Alberta Health using the secure file transfer service. The process for terminating an SFTP account is provided in section 4.6 Terminating an SFTP account.

A user account should have its password reset in response to certain events. The password should be changed if it is accidentally exposed, such as it being accidentally included in an email or present on systems that have been hacked or breached. The password should also be changed whenever support personnel for your external application leave the support team and had knowledge of the password used by the application. The process for resetting a user account password is provided in section 4.5 Resetting an SFTP account password.

4 Tasks

4.1 Installing SFTP software

As an external user, in order to connect to the Alberta Health secure file transfer service using SFTP, you must have SFTP software installed on your computer. The SFTP protocol is implemented by various software packages with varying business models.

Alberta Health is not responsible for the support of SFTP software. The user's organization is responsible.

4.1.1 Installing SFTP with Helpdesk support

- If your organization has a Helpdesk that supports you, you should contact that Helpdesk to determine if SFTP software is already installed on your machine.
- If it is not installed, your organization may have selected a standard SFTP software package, or might have an approved list of SFTP software packages with a preferred option. In that case, you should request to have that software installed on your machine.
- If your organization has not yet selected a standard SFTP software package, and does not permit installation of software packages by users, then a request must be made to select and then install an SFTP software package.
- If your organization does not have a standard SFTP software package, but permits installation of software by users, proceed with the instructions in the next section.
- Your organization's firewall must allow outbound connections on TCP port 22.

4.1.2 Installing SFTP without Helpdesk support

Disclaimer

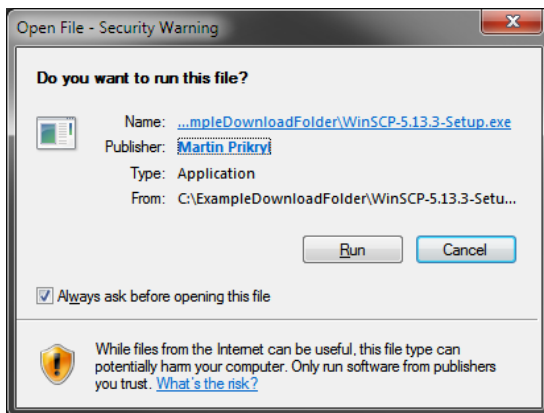
Alberta Health does not supply or support SFTP software.

This section provides instructions on how to install the WinSCP package on Microsoft Windows. However, the choice of WinSCP and these instructions are provided only as an example. At the time of writing, this software package (5.21) was free and open-source, and did not bundle additional software.

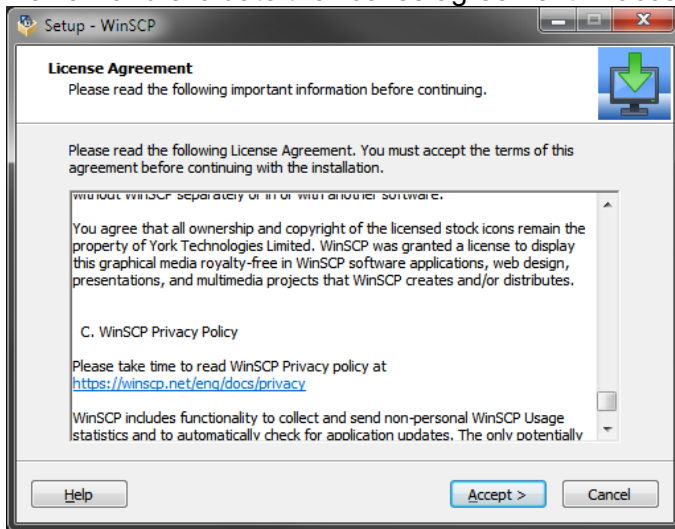
Ultimate responsibility for evaluating software (including software licensing) and supporting software remains with your organization, and not Alberta Health.

Follow these steps to install the WinSCP SFTP package:

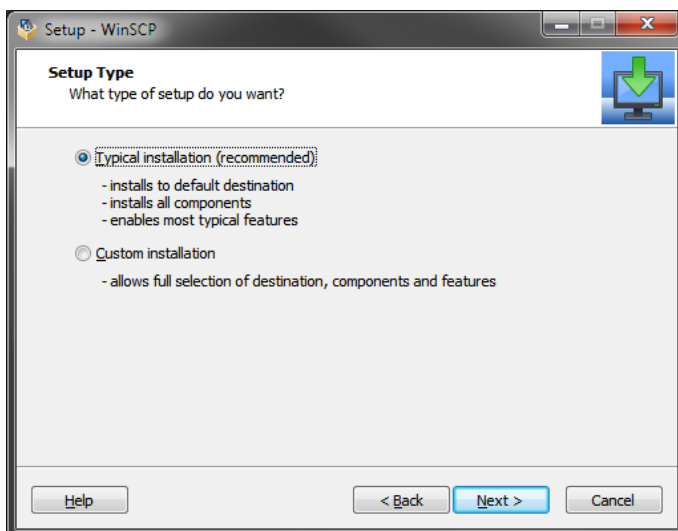
1. Download the software installer from <https://winscp.net/>
2. Launch the WinSCP Setup.exe file.
3. Verify the application signature is as expected, and click “Run”.



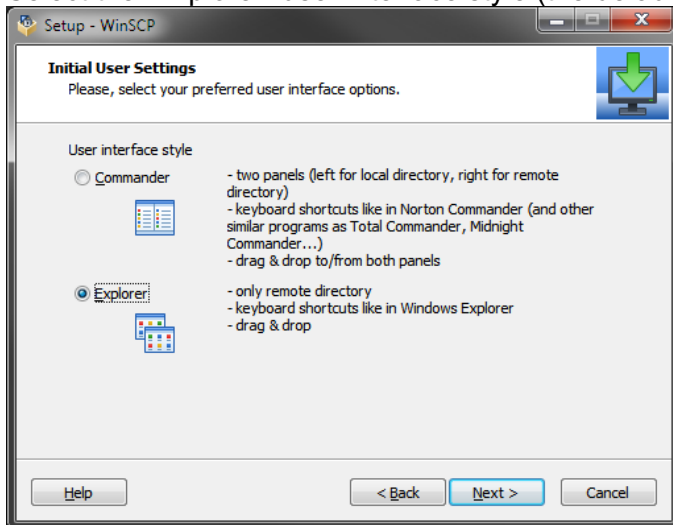
4. Review and evaluate the license agreement. If acceptable, proceed using the “Accept” button.



5. Select a “Typical” installation, and click “Next”.

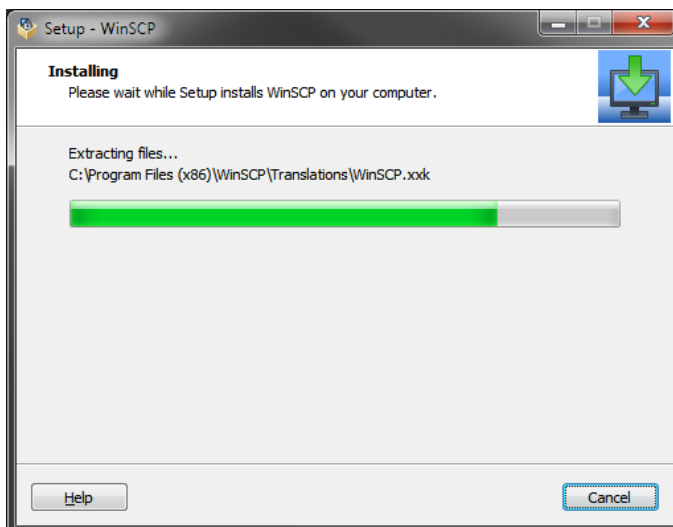


6. Select the “Explorer” user interface style (the default), and click “Next”.



7. Click “Install” to begin the installation process

8. Wait for the installation step to finish



9. Click Finish, which will open a general getting started guide for the software, and launch the program.



Next steps:

- Generating an SSH key pair without Helpdesk support on step 4.2.3
- Creating a SFTP account on step 4.3
- Setup SFTP software without Helpdesk support on step 4.4.2

4.2 Generating an SSH Key Pair

4.2.1 Purpose of an SSH Key Pair

The SFTP protocol is a specific application of the more general Secure Shell (SSH) protocol. One of the common features offered by SSH is authentication. Multiple authentication methods are implemented in SSH, among them passwords and asymmetric encryption keys. Asymmetric encryption relies on a pair of keys. One key is public, and is sent to others. The other key is private, and is kept as a secret from others, just like a password. The public key can be used to encrypt data such that only the holder of the private key can see it, and the public key can be used to validate that data comes from the holder of the private key. To set up the key pairs to work this way, they must be generated and then stored, rather than chosen and remembered (as with password).

To enhance the security of the Alberta Health secure file transfer service, SFTP users must authenticate with both a user id/password and asymmetric encryption key. This provides two-factor authentication: the password is something you know (factor 1), and the private key is something you have (factor 2). This raises the difficulty of a successful attack, as a hacker must obtain multiple pieces of information. However, it also means that if your private key is on one computer, you will only be able to use SFTP on that computer, as you will be missing one of the two factors on other computers.

Remember that your private key is a secret, as is your password. It should not be shared with anyone, including Alberta Health. Your private key should be stored securely on a machine with access controls, and if transferred between machines, the transfer method should be secure.

4.2.2 Generating an SSH Key Pair with Helpdesk Support

This task requires your organization's Helpdesk to have selected and installed SFTP software on your machine, as per section 4.1.1 – Installing SFTP with Helpdesk Support

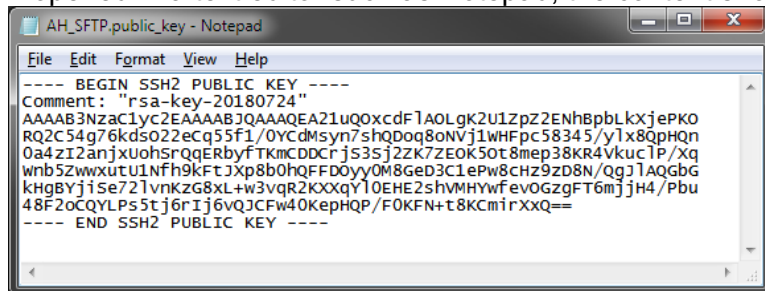
Your organization's Helpdesk will have to assist you in generating an SSH key pair. This may be a service they provide, or it may be that they refer you to the documentation for the SFTP software. Generation of SSH key pairs is a usual feature of SFTP software.

The generation feature should have various adjustable parameters, which should be set as follows:

Option	Required value	Comments
Format	SSH-2	
Algorithm	RSA	May be combined with format.
Key bit size	2048 or larger	
Password / passphrase	required	You can secure your private key with a password, which would be required to use the key. This configuration would mean that two passwords and the private key would be needed to log into Alberta Health SFTP. The two passwords are separate and must not be confused.

The output of the key generation will be two keys, which are usually represented as files. If there is no designated area for saving the keys, you can save the public key to the Desktop with a .public_key extension, and the private key can be saved in an "SFTP Keys" folder under your documents. The public key is often given the .pub extension, although this is also used by Microsoft Publisher.

If opened in a text editor such as Notepad, the content should look like this:



```
----- BEGIN SSH2 PUBLIC KEY -----
Comment: "rsa-key-20180724"
AAAAB3NzaC1yc2EAAAABJQAAAAEA21uQ0xcdf1AOLgk2U1Zpz2ENhBpBLKxjePKO
RQ2C54g76kdsO22ecq55f1/0Ycdmsyn7shQDoq8oNVj1WHFpc58345/y1x8qpHQn
0a4zI2anjxuohSrQqERbyFTkmCDDCrjs3Sj2ZK7ZEOK5ot8mep38KR4vkuc1P/Xq
wnb5ZwwxutU1Nfh9kftJxp8b0hQFFD0yy0M8Ged3C1ePw8chZ9zd8N/QgJ1AqgbG
kHgBYjise7lVnKzG8xL+w3vqR2KXXqY10EHE2shVMHYwfevOGzgFT6mjH4/Pbu
48F2oCQYLPs5tj6rIj6vQJCFw40KepHQP/F0KFN+t8KcmirXxQ==
----- END SSH2 PUBLIC KEY -----
```

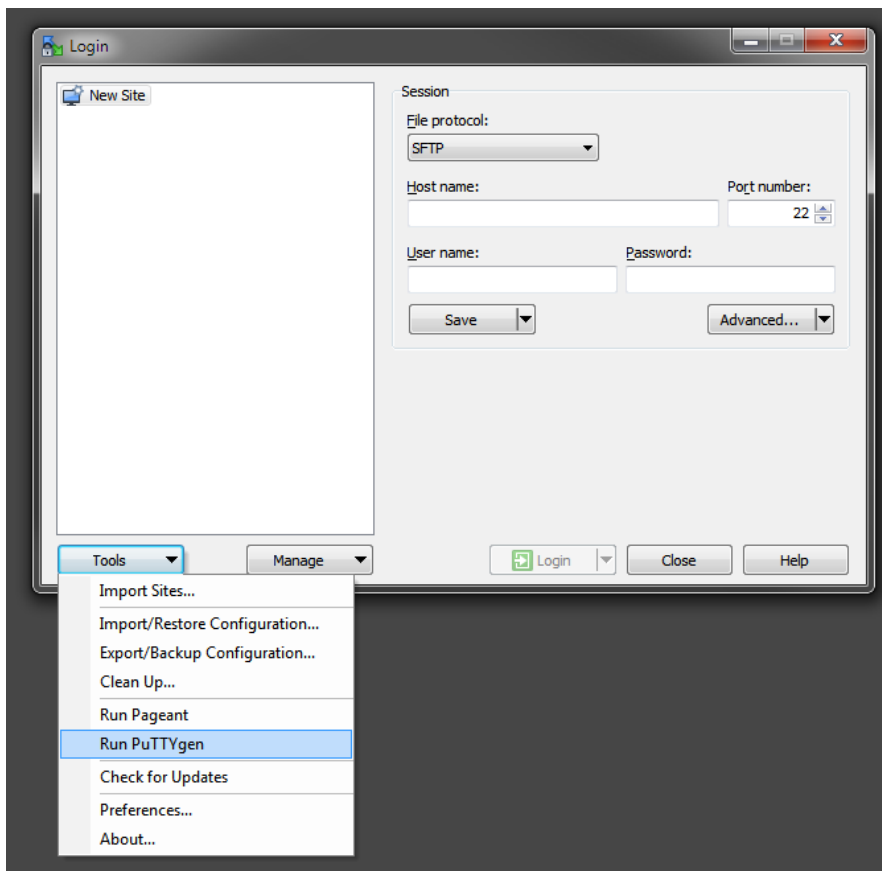
The public key will be needed when creating an SFTP account, as described in section 4.3 – Creating an SFTP account. The private key must be kept secret, including from Alberta Health. It will be used to set up the connection to SFTP, as described in section 4.4.1 – Setting up SFTP Software with Helpdesk Support.

4.2.3 Generating an SSH Key Pair without Helpdesk Support

This section provides a guide to generating an SSH key pair using WinSCP, the software installed in section 4.1.2 – Installing SFTP without Helpdesk support.

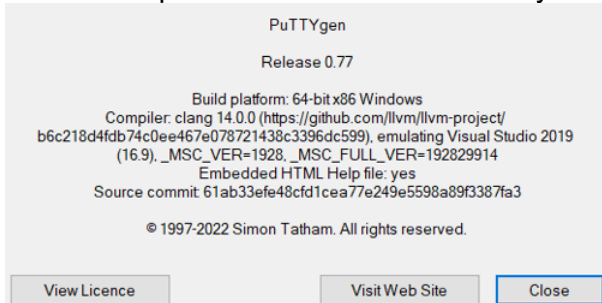
Follow these steps:

1. Open WinSCP, if it is not already running.
2. In the “Tools” menu on the connection dialog, select the “Run PuTTYgen” option. PuTTYgen is a separate component for generating SSH key pairs.

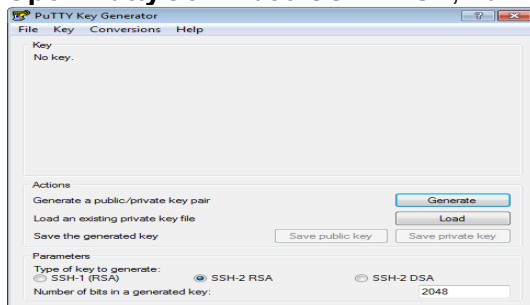


3. Creation of SSH key pairs

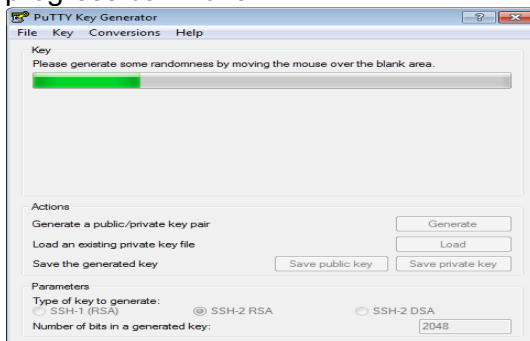
Click on Help – About – to check the PuTTYGen version (use latest version)



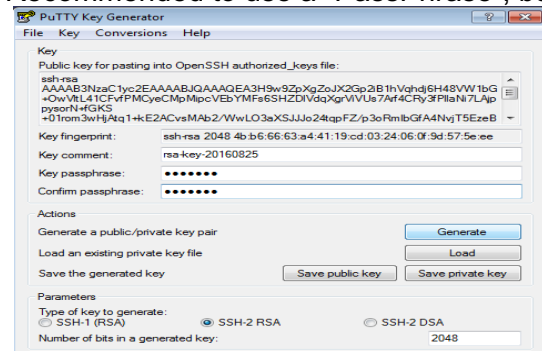
Open PuTTYGen – use SSH2 RSA, number of bits = 2048 as a minimum



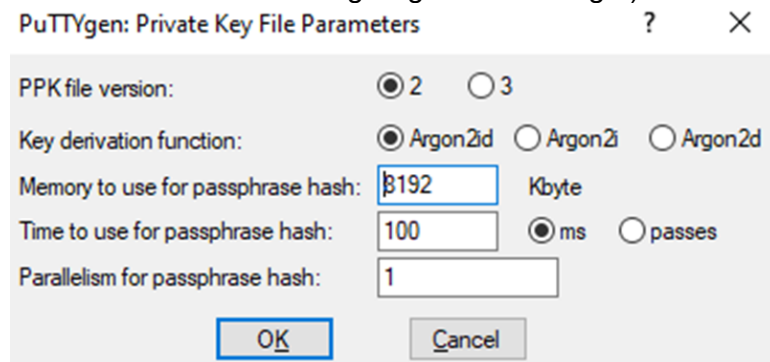
Click “Generate” button, While continuously moving the mouse over the “blank area” you will see the progress bar move:



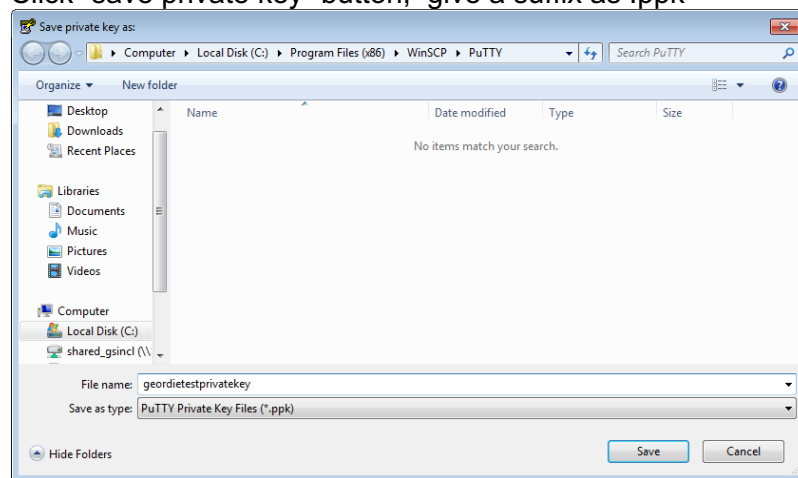
Recommended to use a “PassPhrase”, but not required.



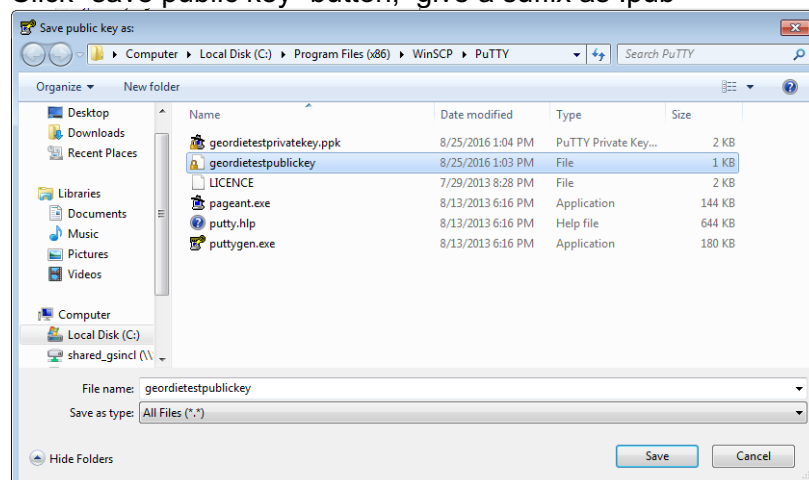
Click on “key” on the file menu, then click on “Parameter for saving key files” – change PPK file version to “2” instead of using the default “3” (only newer version of PuTTYGen has this option, PPK file version 3 will cause issue without giving error message.)



Click “save private key” button, give a suffix as .ppk



Click “save public key” button, give a suffix as .pub



How to submit new public key

- a) For HLINK – please email Health.hlink@gov.ab.ca
- b) For PB – please email Joanne Bain for Kroll (joanne.bain@cgi.com)
- c) For Other Applications – contact your business unit at Alberta Health.

4.3 Creating an SFTP account

A request to create an SFTP account must go through an Alberta Health business owner, as the owner must add additional information and provide their approval for the request.

At a minimum, the following details are required when creating an SFTP account:

- User's first name
- User's middle name
- User's last name
- User's business phone number
- User's email address
- An SSH public key (see section 4.2 – Generating an SSH Key Pair)
- If the SFTP account is for a new user that is being added to an existing submitter group, this detail should be included in the request. (For an explanation of the relationship between Alberta Health business areas, submitters, and external users, see section 2 – File exchange with Alberta Health.

Individual Alberta Health business areas may require additional steps or information outside of the minimum required for the secure file transfer service, and may designate particular contact points for account requests.

Once the appropriate information is provided and the Alberta Health business owner approves, the internal "SFTP User ID Registration" form will be filled out and then processed by Alberta Health.

The GoAnywhere system that implements the Alberta Health secure file transfer service will email the user when the account is created. Two emails are sent: one includes the user ID, and the other includes the password. Only the SFTP interface to GoAnywhere is available outside of Alberta Health, so references to other interfaces (such as the website) in the emails should be ignored.

Business Owner contact

- a) For HLINK – please email HLINK Team, health.hlink@gov.ab.ca
- b) For PB – please email Joanne Bain for Kroll (joanne.bain@cgi.com)
- c) For Other Applications – contact your business unit at Alberta Health.

4.4 Setting up SFTP Software

Once a user id has been created as per section 4.3 – Creating an SFTP account. The software installed as per section 4.1 – Installing SFTP software must be configured to access Alberta Health.

4.4.1 Setting up SFTP Software with Helpdesk Support

This task requires:

- your organization's Helpdesk to have selected and installed SFTP software on your machine, as per section 4.1.1 – Installing SFTP with Helpdesk support.
- your organization's Helpdesk to have created a private key or to have assisted you in the creation of a private key, as per section 4.2.2 – Generating an SSH Key Pair with Helpdesk Support.
- you to have an SFTP user account created, as per section 4.3 – Creating an SFTP account.

Your organization's Helpdesk will have to assist you in configuring the connection in the SFTP software. This may be a service they provide, or it may be that they refer you to the documentation for the SFTP software.

The connection feature will require parameters, some of which may be set beforehand, and some of which may be asked for when connecting.

The parameter values should be set as follows:

Parameter	Value	Comments
Host name	getfile.health.alberta.ca	
Port	22	This should be the default value
User name		This is the identifier of the SFTP user account.
Authentication methods	Password and public key	This parameter may be implicit – for example, if you provide both a password and a private key, both will be used.
Password		This is the password of the SFTP user account.
Private key		This will be the private key file you have previously generated.
Private key password / passphrase		If you opted to secure your private key with a password or pass phrase when the key was saved, then it will be needed as well.

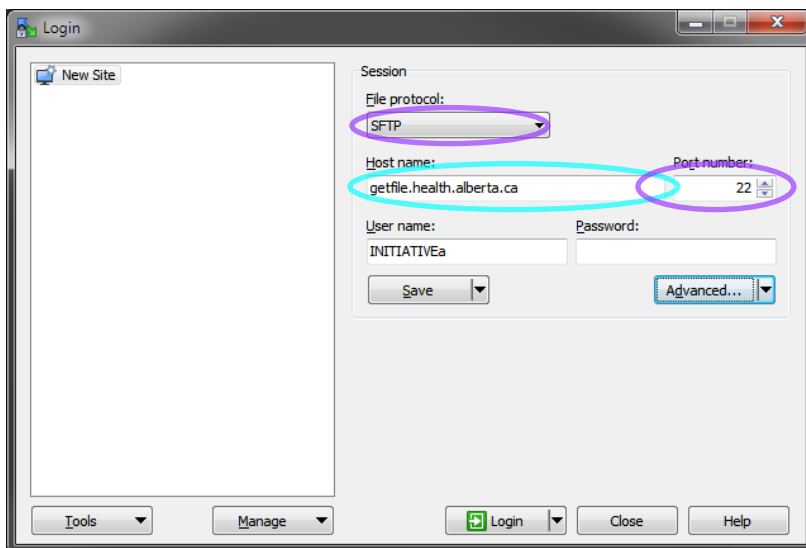
4.4.2 Setting up SFTP Software without Helpdesk Support

This task requires:

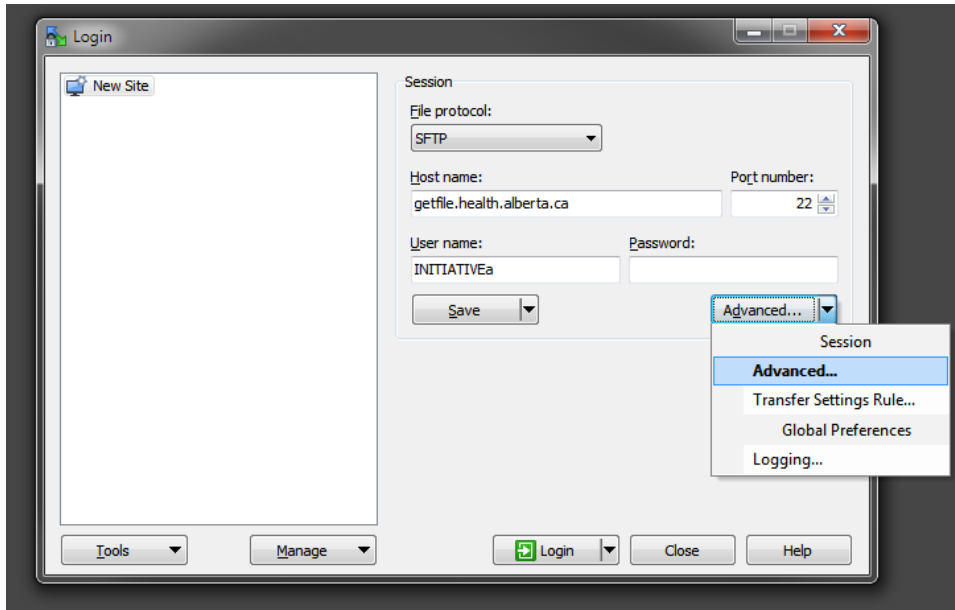
- you to have installed WinSCP on your machine, as per section 4.1.2 – Installing SFTP without Helpdesk support.
- you to have created a private key, as per section 4.2.3 – Generating an SSH Key Pair without Helpdesk Support.
- you to have an SFTP user account created, as per section 4.3 – Creating an SFTP account.

Follow these steps to configure WinSCP to connect to Alberta Health via SFTP:

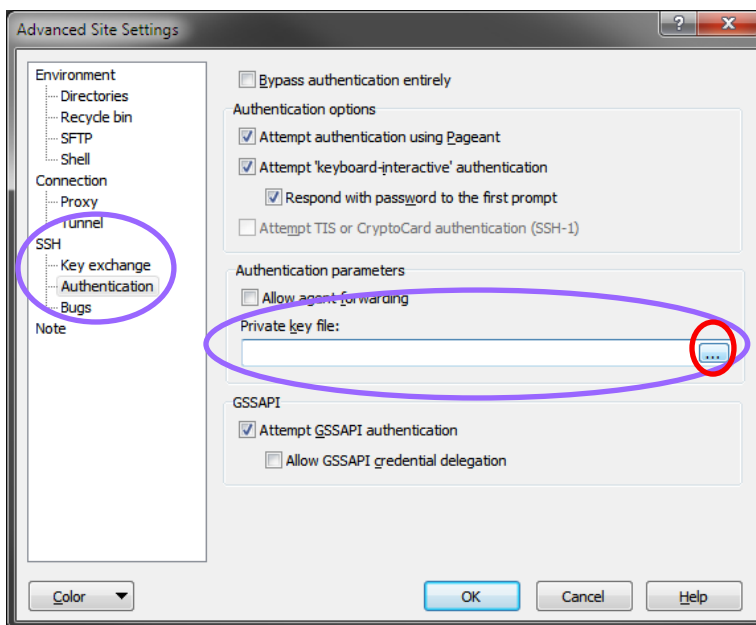
1. Launch WinSCP.
2. Enter `getfile.health.alberta.ca` in the “Host name” field. Enter your user name in the “User name” field. Ensure the “File protocol” is SFTP, and the port number is 22.



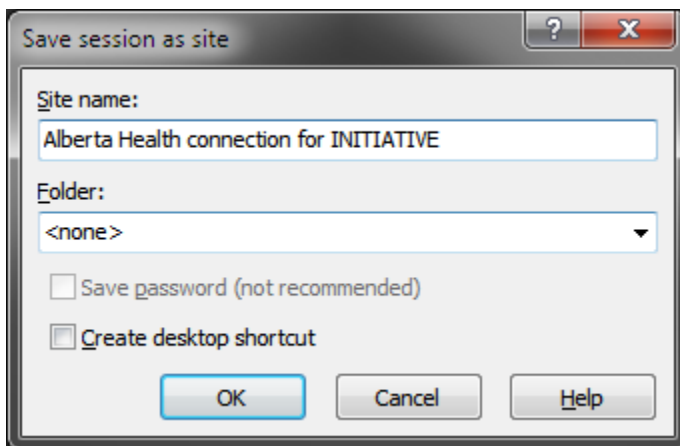
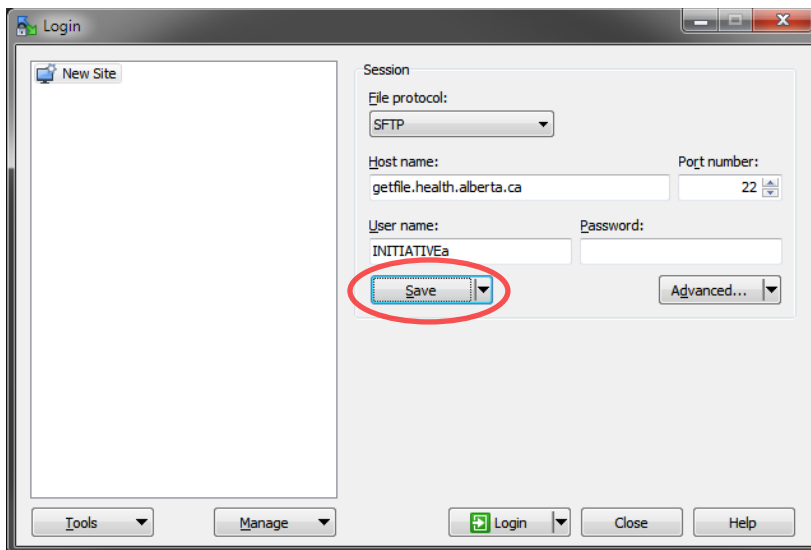
3. Select the “Session > Advanced...” item under the “Advanced...” menu button.



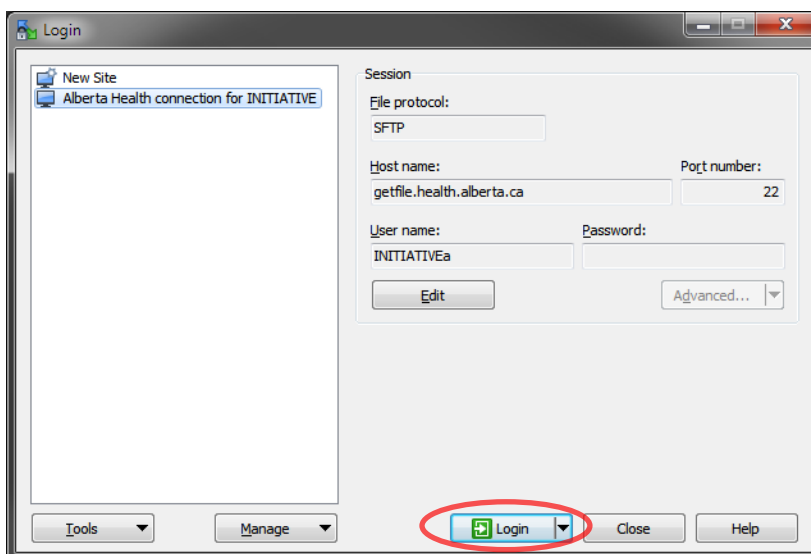
4. Select “SSH > Authentication” in the tree on the left. In the “Private key file” field, click the browse button to navigate to the folder containing the private key file. Click “OK” to exit the dialog”



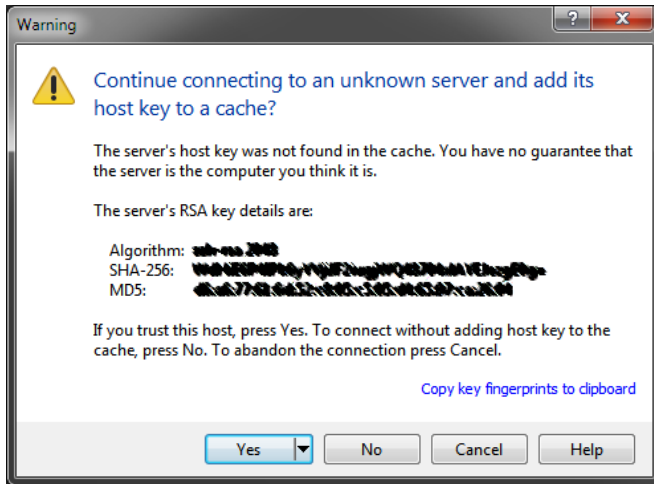
5. Click the Save button. In the next dialog, provide a name for the saved connection information and click “OK”.



6. Click the Login button.

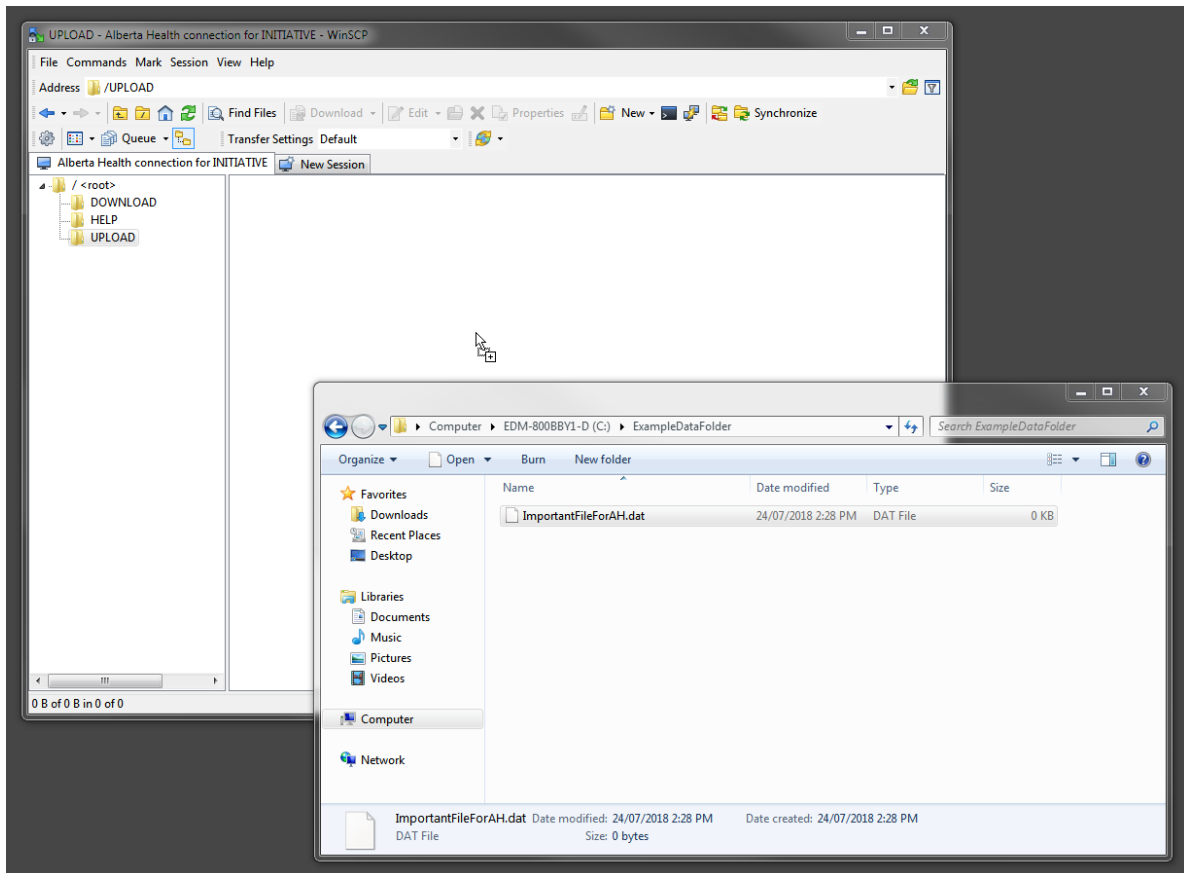


7. As this is the first time the software is connecting with the machine, it will present the following dialog. The expected values for the presented details will change over time and if they do this step will be presented again. Click Yes to proceed.

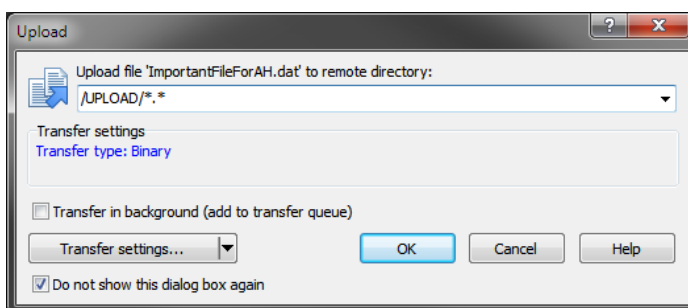


8. You will be prompted for your private key password/passphrase, if you opted to set one up. You will then be prompted for your user password/passphrase.

9. Once connected, you can transfer files by dragging and dropping them to and from Explorer windows.



10. The first time a transfer is made, the following dialog will appear to set up transfer options. The defaults are appropriate. Click “OK”.



4.5 Resetting an SFTP Account Password

Resetting an SFTP account password can either go through Business owner, or AH Provincial Desk.

To reset your SFTP account password:

- a) For HLINK – please email HLINK Team, health.hlink@gov.ab.ca
- b) For PB – please email Joanne Bain for Kroll (joanne.bain@cgi.com)
- c) For Other Applications – contact your business unit at Alberta Health.

Can also submit via Help Desk –

1. Phone the AH Provincial Desk – 780-412-6778 or 1-877-931-1638
2. Tell the operator that you require a password reset for an Alberta Health GoAnywhere account. (GoAnywhere is the name of the system that provides the Alberta Health secure file transfer service.
3. Provide the operator with the account's username.
4. The Service Desk will create a password reset request, and the GoAnywhere system will email the user a new password. Only the SFTP interface to GoAnywhere is available outside of Alberta Health, so references to other interfaces (such as the website) in the emails should be ignored.

4.6 Terminating an SFTP Account

If an SFTP account is compromised or is no longer needed, it should be terminated. A request to terminate an SFTP account must go through an Alberta Health business owner, as the owner must add additional information and set the priority of the request.

Business Owner contact

- a) For HLINK – please email HLINK Team, health.hlink@gov.ab.ca
- b) For PB – please email Joanne Bain for Kroll (joanne.bain@cgi.com)
- c) For Other Applications – contact your business unit at Alberta Health.

A termination request should include the following information:

- The user name.
- Whether the closure should be expedited (for example, in the case of an abrupt employee termination).
- Whether the associated submitter group should also be removed, or if the user is one of many and the submitter group should remain. (For an explanation of the relationship between Alberta Health business areas, submitters, and external users, see section 2 – File exchange with Alberta Health.

Individual Alberta Health business areas may require additional steps or information outside of the minimum required for the secure file transfer service, and may designate particular contact points for account requests.

Once the appropriate information is provided and the Alberta Health business owner approves, the internal “SFTP User ID Registration” form will be filled out and then processed by Alberta Health to terminate the account.

5 Troubleshooting

If need help for GoAnywhere, please contact AH Provincial Service Desk. 780-412-6778 or 1-877-931-1638.